

Circular Rendezvous Simulation

A probabilistic simulation of two walkers moving on a circular track under uncertainty.

Problem Statement

Two walkers move on a circular track of circumference C .

Each walker:

- starts at a random position,
- chooses a random direction,
- may reverse direction probabilistically,
- cannot reverse until traveling a minimum persistence distance D .

The objective is to study:

- Expected meeting time $E[T]$
- Effect of reversal probability p
- Ballistic vs diffusive behavior
- Persistent random walks
- Symmetry breaking in rendezvous search

Mathematical Ideas

This project combines:

- Rendezvous search theory
- Persistent random walks
- Diffusion processes
- Monte Carlo simulation
- Stochastic motion on compact geometry

Parameters

Parameter	Meaning
C	Circumference of circle
p	Reversal probability
D	Minimum distance before turning
$E[T]$	Expected meeting time

Example Questions

- How does $E[T]$ vary with p ?
 - Is there an optimal persistence distance?
 - When does motion become ballistic vs diffusive?
 - What happens if turning distances follow a Gaussian distribution?
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Example Plot

$E[T]$ vs p

 $E[T]$ vs p

$E[T]$ vs D

 $E[T]$ vs D

Run

Install dependencies:

```
pip install -r requirements.txt
```